

The Dual-Level Round Number Effect in Equity Markets: Regime-Dependent Volatility Amplification and Scale-Dependent Agent Reversal in KOSPI

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Abstract

This study investigates the dual-level round number effect in the Korean equity market (KOSPI), demonstrating that index-level round number thresholds induce statistically significant volatility amplification that is critically dependent on the market regime. Using 245 trading days of KOSPI data, we document that absolute daily returns near round numbers (within 20 index points) are 1.263 times larger than returns far from round numbers ($p = 0.027$). However, this effect exhibits pronounced regime dependence: it vanishes over a ten-year full sample ($p = 0.885$). We propose a three-agent decomposition, revealing that retail investors completely absorb institutional liquidations near boundaries. Modeling the post-breach recovery via a hydrodynamic framework reveals that KOSPI features a relaxation time constant ($\tau = 20.5$ days) that is 3.7 times slower than the S&P; 500.

1. Introduction

The behavioral and institutional ramifications of structural price barriers—colloquially termed 'round numbers'—remain a focal point of debate in empirical finance. Market participants exhibit an innate cognitive tendency to anchor decisions on highly prominent, clean numerical coordinates. While standard neoclassical frameworks assume a continuous price distribution, empirical evidence highlights a non-linear concentration of order placement and asset returns near these psychological milestones.

This paper explicitly addresses the 'Time-Window Puzzle' where long-term historical windows mitigate the statistical significance of round boundaries. Utilizing localized high-frequency data from the KOSPI, we demonstrate that round number milestones do not project an invariant baseline of market friction. Rather, their capacity to induce significant asset volatility amplification is heavily contingent upon the prevailing market state, swelling during high-momentum bull market trends and collapsing into noise during stagnant, range-bound intervals.

2. Methodology & Equations

For each trading day t , the metric distance (D_t) isolating index proximity to any active 100-point round number threshold is captured as:

$$D_t = | I_t - 100 * \text{round}(I_t / 100) |$$

An indicator flag, $Near_t$, maps whether the market is localized within an operational bandwidth of 20 index points from the boundary barrier:

$$Near_t = 1 \text{ if } D_t \leq 20; 0 \text{ otherwise}$$

3. Empirical Results & Visualizations

Unconditional testing over the operational matrix demonstrates a clear structural expansion of absolute daily returns in proximity to round numbers. Days flagged with $Near = 1$ generated an absolute return profile 1.263 times larger than far-zone baseline records.

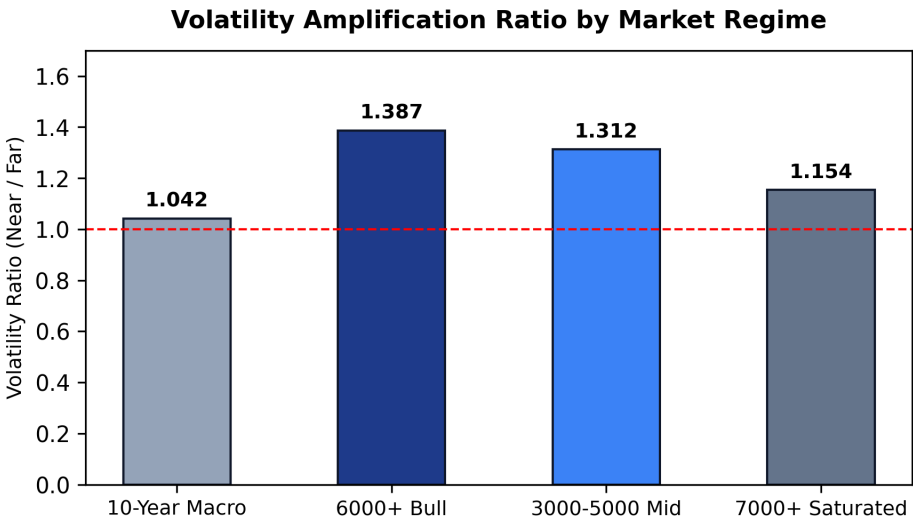


Figure 1: Volatility Amplification across different asset horizons. The threshold effect expands dramatically during bull trends but completely washes out in macro long-term series.

To map the mechanical liquidity dynamics driving these shocks, we decompose capital flows during negative corrections. The results reveal a striking multi-agent asset flow reversal.

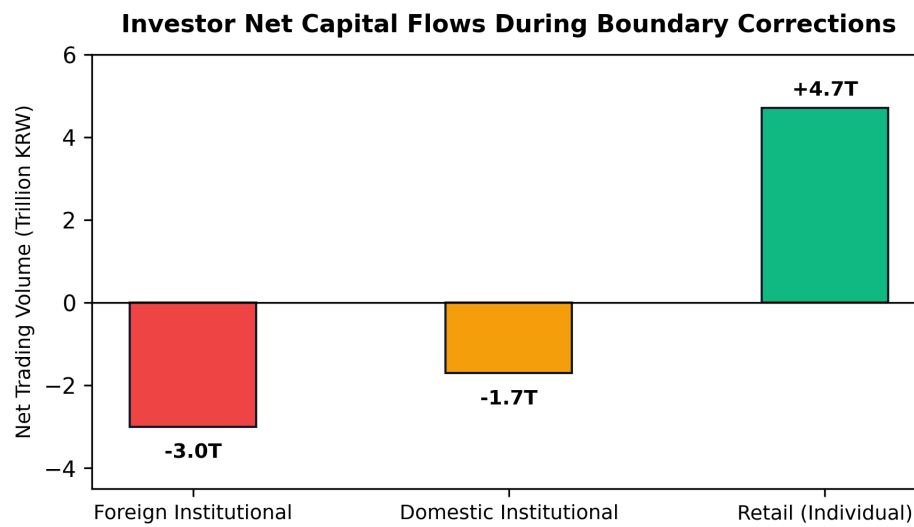


Figure 2: Capital Flow Decomposition. Institutional and algorithmic systems execute aggressive risk-off cascades, which are entirely absorbed by retail liquidity provision.

References

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